BEHIND THE SCREEN

The Hidden Cost: Electricity, Cooling, and Resources

HM216 · Corrected research report for an educational website Revised 24 October 2026

Evidence status and scope

This revision corrects and qualifies the supplied Antigravity report. “Verified” means the cited source directly supports the claim as worded; it does not mean every related interpretation is proven. Forecasts are labelled as forecasts, corporate disclosures as company-reported data, and model outputs as illustrative calculations.

Detailed claims in the original report that lacked a directly identifiable source or reproducible method are not presented as verified. The source-links appendix provides direct links to the principal sources used and to additional source material for further checking.

Executive summary

Digital services rely on physical infrastructure: data centres, networking equipment, electricity grids, cooling systems, and the manufacture and eventual disposal of computing hardware. These systems consume electricity and, depending on cooling design and location, can also consume substantial amounts of water. Hardware manufacturing and end-of-life management add environmental impacts beyond the electricity used during operation.

The International Energy Agency (IEA) estimated that data centres, artificial intelligence (AI), and cryptocurrency together used about 460 terawatt-hours (TWh) of electricity worldwide in 2022, nearly 2% of global electricity demand. In its January 2024 report, the IEA modelled a 2026 range of approximately 620–1,050 TWh, with a base case just above 800 TWh. These are a historical estimate and a forecast published in 2024—not measured results for 2026. The combined category includes cryptocurrency and AI, and excludes electricity used by data-transmission networks [1].

Masanet and colleagues estimated that global data-centre electricity use increased from about 194 TWh in 2010 to 205 TWh in 2018 (about 6%), while compute instances increased by 550%. This is evidence of historical efficiency gains, not proof that energy demand will always remain decoupled from computing growth [2].

Environmental impact is not captured by a single metric. Power Usage Effectiveness (PUE) describes facility energy relative to IT energy; Water Usage Effectiveness (WUE) relates site water consumption to IT energy; and greenhouse-gas inventories depend on system boundaries and accounting methods. A responsible educational website should show these metrics with definitions, dates, assumptions, and uncertainty rather than assigning a universal electricity or carbon value to every click or AI prompt.

1. Electricity demand and efficiency

1.1 Global estimate and forecast

The IEA’s Electricity 2024 report estimated 460 TWh for the combined data-centre, AI, and cryptocurrency category in 2022. Its 2026 scenarios ranged from 620 to 1,050 TWh, with a base case slightly above 800 TWh. The IEA noted that computing accounted for about 40% of data-centre electricity demand and cooling for about another 40%, with other associated IT equipment making up the remainder [1].

Do not describe the 620–1,050 TWh range as the amount actually consumed in 2026. It was a forecast produced in 2024. To report observed 2026 consumption, use a newer dated source and verify that its scope is comparable.

1.2 Historical efficiency gains

Masanet et al. estimated global data-centre energy use at about 194 TWh in 2010 and 205 TWh in 2018, an increase of roughly 6%, while compute instances increased by 550%. The authors discussed improvements in server and storage efficiency and structural changes in the data-centre sector [2].

This result shows that technology and operational changes can reduce energy per unit of computing service. It does not establish that future electricity use will grow slowly or that all efficiency opportunities have been exhausted.

2. Cooling and facility efficiency

Computing equipment converts nearly all of the electricity it uses into heat that must ultimately be removed from the facility. Cooling may use fans, pumps, chillers, heat exchangers, evaporative systems, or liquid cooling close to the chips. The best design depends on server heat density, climate, water availability, electricity supply, reliability needs, and local engineering constraints.

2.1 PUE: what it does and does not say

Power Usage Effectiveness is total data-centre facility energy divided by IT-equipment energy. A PUE of 1.50 means that for every 1 kWh consumed by IT equipment, the facility uses 1.50 kWh in total, including overhead. PUE is a ratio, not a direct measure of carbon emissions, water use, or overall sustainability.

Uptime Institute reported a global average PUE of 1.56 in its 2024 survey, compared with 1.58 in 2023. These are survey results, not universal values for every facility [3].

The original report's proposed PUE overhead ranges for DX cooling, chilled-water systems, air-side economization, and direct-to-chip liquid cooling are not retained as universal coefficients. Their values depend on climate, load, equipment, and system boundary, and a sufficiently specific source was not provided for each range.

3. Water use and local impacts

3.1 Withdrawal versus consumption

Water withdrawal is the volume taken from a source; water consumption is the portion not returned to the same local water system in a usable form, for example because it evaporates. These terms should not be used interchangeably. Evaporative cooling can reduce electricity used for cooling in suitable conditions, but it can increase on-site water consumption. Dry cooling can reduce on-site water use but may require more electricity in some conditions.

WUE (Water Usage Effectiveness) is commonly expressed as litres of site water consumed per kWh of IT energy. A value is meaningful only when its reporting boundary and period are stated. A reported site WUE of zero would mean no site water consumption within the metric's boundary; it would not mean that electricity supply or equipment manufacturing has no water footprint.

3.2 Google data-centre water reporting

Google's 2024 Environmental Report, covering 2023, reported approximately 6.1 billion US gallons of water consumption by its data centres in 2023 (about 23.1 billion litres). This is a company-reported figure and should be interpreted within the report's stated boundary [4].

3.3 The Dalles, Oregon case study

The Dalles is a useful case study because local water use can be examined using municipal records and local reporting. Oregon Public Broadcasting reported, citing city-provided records, that Google used 383.9 million gallons in The Dalles in 2023 and 434.4 million gallons in 2024 [5]. State the year and source whenever using these numbers.

A large customer's measured water use establishes demand from that customer; by itself it does not prove that the customer caused regional drought or aquifer decline. Claims about the share of city supply, court proceedings, settlement terms, or aquifer impacts require a direct municipal record or reliable report for each claim.

4. Carbon emissions and accounting boundaries

Operational emissions depend on electricity consumed and the emissions intensity of the electricity supply. For a simplified estimate: emissions (kg CO■e) = electricity consumed (kWh) × grid intensity (kg CO■e/kWh). State whether the calculation covers IT equipment alone or the whole facility, which grid factor is used, and whether it is location-based or market-based.

Corporate inventories distinguish Scope 1 (direct emissions from owned or controlled sources), Scope 2 (emissions associated with purchased electricity, steam, heat, and cooling), and Scope 3 (other value-chain emissions). Location-based and market-based Scope 2 figures can differ because they use different accounting methods. A

market-based figure should not be described as the physical emissions intensity of the local grid.

The exact Microsoft FY2023 figures in the supplied report are not reproduced here because the cited document/page references were not independently matched to an accessible primary table during this review. Restore them only after checking the original company report and confirming fiscal year, units, scope, and accounting method [7].

5. Hardware lifecycle and electronic waste

The Global E-waste Monitor 2024, produced by UNITAR and ITU, estimated that the world generated 62 million tonnes of e-waste in 2022. Only 22.3% was documented as formally collected and recycled in an environmentally sound manner. This is a global total for discarded electrical and electronic equipment; it must not be attributed specifically to data centres [6].

Data-centre equipment is part of the broader electronics lifecycle. Manufacturing servers and accelerators requires materials, energy, and complex supply chains; retirement creates opportunities and risks related to repair, reuse, refurbishment, recycling, and hazardous waste handling. The claim that embodied emissions account for 70–80% of lifecycle emissions on low-carbon grids is not stated as universal because its exact scope and conditions need checking.

6. Recommended educational simulator

Use a transparent facility-level model rather than assigning a supposedly exact physical cost to one click, search, video, or AI prompt. Let users change IT load, PUE, WUE, and grid emissions intensity. State clearly that the result is an illustrative annual facility scenario, not a measurement of a user’s personal internet activity.

6.1 Equations

Annual IT electricity (kWh) = IT load (kW) × operating hours per year.

Annual facility electricity (kWh) = annual IT electricity × PUE.

Estimated site water consumption (litres) = annual IT electricity (kWh) × site WUE (L/kWh).

Estimated operational emissions (kg CO■e) = annual facility electricity (kWh) × grid factor (kg CO■e/kWh).

Boundary note: WUE is normalized to IT energy, while the emissions calculation uses total facility electricity. Do not multiply water by PUE unless the chosen water coefficient is explicitly defined on a facility-energy basis.

6.2 Worked example

Assume a constant 10 MW IT load for 8,760 hours/year, PUE = 1.15, site WUE = 1.8 L/kWh, and a grid emissions factor of 360 g CO■e/kWh. These are illustrative inputs, not universal industry averages.

Annual IT energy = 10,000 kW × 8,760 h = 87,600,000 kWh (87,600 MWh).

Annual facility electricity = 87,600,000 × 1.15 = 100,740,000 kWh (100,740 MWh).

Estimated site water consumption = 87,600,000 × 1.8 = 157,680,000 litres.

Estimated operational emissions = 100,740,000 × 0.360 kg CO■e/kWh = 36,266,400 kg CO■e (about 36,266 tonnes).

These calculations are arithmetically consistent with the stated inputs. For a real-world case study, the WUE and grid factor must be sourced for the chosen facility and location.

7. Claims removed, downgraded, or awaiting verification

8. Conclusion

The internet’s environmental footprint is real but multi-dimensional. Electricity demand depends on computing workloads and facility design; cooling choices can shift impacts between electricity and water; emissions depend on electricity supply and accounting boundaries; and hardware production and disposal create lifecycle impacts. The most reliable public explanation separates measured data from forecasts, avoids attributing global e-waste totals

specifically to data centres, and presents any simulator as a transparent scenario model. Claims that could not be independently confirmed are explicitly marked rather than silently repeated.

References and direct source links

[1] International Energy Agency (IEA). (2024). Electricity 2024: Analysis and forecast to 2026. Source: https://www.iea.org/reports/electricity-2024 Additional source / PDF: https://iea.blob.core.windows.net/assets/6b2fd954-2017-408e-bf08-952fdd62118a/Electricity2024-Analysisandforecastto2026.pdf

[2] Masanet, E., Shehabi, A., Lei, N., Smith, S., & Koomey, J. (2020). Recalibrating global data center energy-use estimates. Science, 367(6481), 984–986. Source: https://doi.org/10.1126/science.aba3758 Additional source / PDF: https://seta.lbl.gov/publications/recalibrating-global-data-center

[3] Uptime Institute. (2024). Uptime Institute Global Data Center Survey 2024. Source: https://intelligence.uptimeinstitute.com/resource/uptime-institute-global-data-center-survey-2024 Additional source / PDF: https://datacenter.uptimeinstitute.com/rs/711-RIA-145/images/2024.GlobalDataCenterSurvey.Report.pdf

[4] Google. (2024). 2024 Environmental Report (reports on 2023). Source: https://blog.google/company-news/outreach-and-initiatives/sustainability/2024-environmental-report/ Additional source / PDF: https://sustainability.google/reports/

[5] Oregon Public Broadcasting (OPB). (23 January 2026). The Dalles' mayor called OPB's data center story inaccurate. Here are the facts. Source: https://www.opb.org/article/2026/01/23/the-dalles-mayor-data-center-google/

[6] UNITAR & International Telecommunication Union (ITU). (2024). The Global E-waste Monitor 2024. Source: https://www.itu.int/en/ITU-D/Environment/Pages/Publications/The-Global-E-Waste-Monitor-2024.aspx Additional source / PDF: https://www.who.int/news-room/fact-sheets/detail/electronic-waste-%28e-waste%29

[7] Microsoft. Sustainability reports and data. Specific FY2023 figures from the supplied report remain pending verification against the original tables. Source: https://www.microsoft.com/en-us/corporate-responsibility/sustainability/reports

[8] Luccioni, A. S., et al. (2023). Power Hungry Processing: Watts Driving the Cost of AI Deployment. Source: https://arxiv.org/abs/2311.16863

[9] Kamiya, G. (2020). The carbon footprint of streaming video: fact-checking the headlines. International Energy Agency. Source: https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines

[10] Lawrence Berkeley National Laboratory. United States Data Center Energy Usage Report. Source: https://eta.lbl.gov/publications/united-states-data-center-energy

[11] HM216 course source: Behind the Screen — Group Briefing Report, pp. 2, 4–5, 7. Uploaded course PDF; this is the assignment specification, not evidence for external statistics. Source: Uploaded by user in this conversation.

Submission note

The HM216 briefing asks for direct links to original, credible sources, dated and geographically scoped statistics, a documented example, and explicit limitations. This report follows that structure. Before submission, keep claims marked unverified excluded from the website unless you obtain a source that directly supports them. The course briefing PDF is cited as the assignment specification, not as evidence for the environmental statistics.

Claim from supplied report	Status	Reason / action needed
AI energy per prompt: "1–6 Wh per generation" vs "1–6 Wh per 1,000 queries"	Unverified / inconsistent	Contradictory units. Do not publish without exact model, workload, hardware, and methodology.
Fixed energy figures for a search, AI prompt, or video session	Not used as universal facts	Estimates vary by system boundary, device, network, model, and workload.
Cooling-specific PUE overhead ranges	Unverified	No reproducible source supplied for each range; performance is site-specific.

Claim from supplied report	Status	Reason / action needed
WUE ranges and universal dry-cooling WUE of zero	Qualified	Metric boundary matters; zero site WUE does not mean zero lifecycle water footprint.
Microsoft FY2023 emissions and “96.1%” Scope 3 share	Pending primary-source match	Check original report table, year, units, Scope 2 method, and denominator.
Embodied carbon equals 70–80% of lifecycle emissions	Not stated as universal	Requires exact study boundary, server type, lifetime, utilization, and electricity mix.
Efficiency reserves exhausted or future demand must rise linearly	Removed	Overly strong extrapolation not established by historical efficiency study.
The Dalles court case, legal fees, agreement, and aquifer claims	Not repeated as verified facts	Require exact primary records or reliable article citations for each claim.
“14 times” carbon difference between accounting methods	Removed pending calculation/source	Needs specified company, year, boundary, and comparable Scope 2 values.